ERES INSTITUTE FOR NEW AGE CYBERNETICS

SECURITY, PRIVACY AND TRUST

[DRAFT]

A Unified Specification for Identity Integrity, Biometric Sovereignty, Data-Chain Provenance, and Systemic Trust Architecture within the New Age Cybernetics Framework

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Integrating: IDIPITIS · FAVORS · CBGMODD · Gracechain · SECUIR · NBERS

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DRAFT DOCUMENT — Subject to revision. This specification establishes the architectural intent and formal structure of the ERES Security, Privacy and Trust framework. Final normative status is contingent upon completion of implementation annexes and MIEVM ensemble validation.

PREAMBLE

The Security, Privacy and Trust specification (hereinafter “SPT”) constitutes the unified protective architecture of the ERES Institute for New Age Cybernetics. It brings into formal coherence six previously independent but deeply interlocking subsystems of the New Age Cybernetics (NAC) framework: IDIPITIS (identity and security clearance), FAVORS (biometric sovereignty and verification), CBGMODD (data-chain integrity and provenance), Gracechain (trust-ledger and meritocratic credentialing), SECUIR (physical and digital infrastructure protection), and NBERS (national biometric enrollment and rights scaffolding). Where prior ERES publications have specified each subsystem individually, the SPT document articulates the connective tissue—the unified threat model, the shared trust assumptions, the cross-system dependency graph, and the integrated governance logic that makes these six subsystems function as a single protective organism.

The need for this unification is not merely organizational. Security, privacy, and trust are not three separate concerns to be addressed by three separate policy instruments. They form a triune relationship—each is a precondition for the others, each is degraded when the others fail, and each achieves its fullest expression only when the other two are simultaneously healthy. A system with strong security but weak privacy becomes a surveillance apparatus. A system with strong privacy but weak trust becomes an opacity engine. A system with strong trust but weak security becomes a target. The SPT specification is designed to make all three of these failure modes structurally impossible within the NAC framework.

This architecture is morally vectored. Consistent with the ERES epistemology, the SPT does not aspire to neutral technical adequacy. It is oriented toward an explicit end-state in which every individual possesses sovereign control over their identity and biometric data, in which every institution operates under conditions of radical transparency, and in which systemic trust is not a sentiment but a measurable, auditable, and continuously validated property of the governance infrastructure itself.

1. FOUNDATIONAL DEFINITIONS

The following definitions govern the interpretation of all provisions within this specification. Where these definitions conflict with colloquial or industry-standard usage, the SPT definitions prevail within the NAC context.

1.1 Security

Security is the condition in which a system's integrity boundaries are maintained against all classes of adversarial action—unauthorized access, data exfiltration, identity impersonation, infrastructure compromise, and epistemic manipulation. Within the NAC framework, security is not a static perimeter but a dynamic, self-healing property maintained through continuous cybernetic feedback. Security is measured via the BERA Adaptive Resilience Index (ARI) and is operationalized through IDIPITIS clearance protocols and SECUIR infrastructure provisions.

1.2 Privacy

Privacy is the sovereign right of every individual to determine the scope, duration, audience, and purpose of disclosure for all personally identifying information, biometric data, behavioral telemetry, and deliberative reasoning. Privacy is not the absence of data collection but the presence of enforceable consent architecture. Within NAC, privacy is operationalized through FAVORS biometric sovereignty provisions and CBGMODD data-chain provenance tracking, and is measured via the BERA Resonant Harmony Cycle (RHC) as a community-level indicator of informational self-determination.

1.3 Trust

Trust is a systemic property—not a subjective sentiment—that emerges when security and privacy are simultaneously maintained over time, and when the institutions operating within the system demonstrate verifiable adherence to stated commitments. Trust within NAC is not assumed; it is continuously earned, measured, and validated. It is operationalized through Gracechain credentialing, quantified via the BERA Resonant Continuity Index (RCI = P_Ω_norm × ARI_sys × VibConst), and governed by the transparency provisions of the ERES epistemology.

1.4 The SPT Triad Integrity Condition

The SPT specification introduces a formal integrity condition: Security, Privacy, and Trust are considered to be in a state of triad integrity when and only when all three of the following conditions are simultaneously satisfied:

1. **ARI ≥ ARI_floor(tier)** — Security is maintained at or above the minimum Adaptive Resilience Index for the applicable GSSG tier.
2. **RHC ∈ [μ – 1σ, μ + 1σ]** — Privacy conditions maintain community Resonant Harmony within one standard deviation of the rolling mean.
3. **RCI ≥ 0.7 × RCI_baseline** — Systemic trust has not degraded below 70% of the established baseline for the domain in question.

When any one condition fails, the SPT architecture enters a degraded state and mandatory remediation procedures are triggered as specified in Section 9.

2. UNIFIED THREAT MODEL

The SPT specification operates against a unified threat model that recognizes five classes of adversarial action. These classes are not mutually exclusive; sophisticated attacks typically span multiple classes simultaneously, which is precisely why a unified protective architecture is required.

Class	Designation	Description and Scope
I	**IDENTITY COMPROMISE**	Unauthorized creation, duplication, impersonation, or degradation of identity credentials. Includes synthetic identity generation, biometric spoofing, credential theft, and clearance escalation attacks against IDIPITIS. Countermeasure domain: IDIPITIS + FAVORS.
II	**DATA EXFILTRATION**	Unauthorized extraction, duplication, or transmission of protected data from any NAC-governed system. Includes bulk data harvesting, side-channel leakage, coerced disclosure, and supply-chain interception. Countermeasure domain: CBGMODD + SECUIR.
III	**TRUST EROSION**	Deliberate or systemic actions that degrade institutional trustworthiness without necessarily breaching security or privacy perimeters. Includes disinformation campaigns, algorithmic bias injection, selective transparency manipulation, and credential inflation on Gracechain. Countermeasure domain: Gracechain + MIEVM.
IV	**INFRASTRUCTURE DISRUPTION**	Physical or digital disruption of the systems upon which security, privacy, and trust depend. Includes VERTECA energy grid attacks, SECUIR facility compromise, communication backbone severance, and distributed denial-of-service against SOMT/GAIA coordination channels. Countermeasure domain: SECUIR + VERTECA.
V	**EPISTEMIC MANIPULATION**	The most sophisticated and dangerous threat class: attacks against the knowledge-production and decision-making processes themselves. Includes adversarial corruption of MIEVM validation instruments, poisoning of BERA index inputs, manipulation of SCALULAR certification assessments, and insertion of false premises into Public Reasoning Archives. Countermeasure domain: MIEVM + CBGMODD + PlayNAC.

3. IDIPITIS: IDENTITY INTEGRITY AND SECURITY CLEARANCE

IDIPITIS is the identity and security clearance subsystem of the NAC framework. Within the SPT architecture, it serves as the primary boundary-enforcement mechanism—the system that determines who is permitted to access what, under which conditions, and with what degree of accountability.

3.1 Identity Lifecycle

Every identity within the NAC framework passes through a formally defined lifecycle managed by IDIPITIS:

- **Genesis.** Initial identity creation through NBERS enrollment. The individual provides foundational biometric data (processed through FAVORS), receives an SSSC Tier 1 credential via SCALULAR, and is assigned a unique IDIPITIS identity vector. No identity may be created without a verifiable biological person; synthetic or institutional identities are classified separately and carry distinct privilege sets.
- **Maturation.** Progressive credential accumulation through EarnedPath milestones, SCALULAR certifications (SSHP, SSLA, SSPS, SSST), and Gracechain merit accrual. Each credential layer expands the individual's access privileges and simultaneously increases their accountability surface.
- **Active Operation.** The steady-state condition in which an identity is continuously validated against FAVORS biometric checkpoints, BERA indices are maintained within acceptable ranges, and the individual exercises their full credential-authorized access rights.
- **Suspension.** Temporary revocation of specified access privileges triggered by BERA index degradation, security incident investigation, or voluntary self-suspension. Suspension is non-punitive by default and is designed to protect both the individual and the system during periods of elevated risk.
- **Archival.** Terminal state in which an identity is no longer actively operated but is preserved in the permanent record for historical continuity, estate adjudication, and generational knowledge transfer. Archival identities retain read-only access to their own Public Reasoning Archive contributions.

3.2 Security Clearance Architecture

IDIPITIS implements a graduated clearance architecture that maps directly to the GSSG tier system. Clearance is not a binary gate but a continuous variable—an individual's effective clearance at any given moment is a function of their credential stack, their current BERA index values, their biometric verification freshness, and the sensitivity classification of the resource being accessed.

The clearance function is expressed as:

```
CLR_eff = f(CRED_stack, ARI_current, BIO_fresh, RES_class)
```

Where CLR_eff is the effective clearance level, CRED_stack is the cumulative credential weight from SCALULAR and Gracechain, ARI_current is the individual's real-time Adaptive Resilience Index, BIO_fresh is the time-decay function of the most recent FAVORS biometric verification, and RES_class is the sensitivity classification of the target resource. Access is granted when CLR_eff exceeds the resource's minimum clearance threshold; otherwise, the system returns a denial with a transparent explanation of which component(s) fell short.

4. FAVORS: BIOMETRIC SOVEREIGNTY

FAVORS is the biometric subsystem of NAC. Its fundamental design principle is biometric sovereignty: the proposition that an individual's biometric data—fingerprints, retinal patterns, gait signatures, voiceprints, genomic markers—belong to that individual in the same ontological category as their body belongs to them. Biometric data is not "collected" by the system; it is "entrusted" by the individual under conditions they define and can revoke.

4.1 Consent Architecture

FAVORS implements a seven-layer consent architecture governing all biometric data interactions:

Layer	Designation	Governance
1	**COLLECTION**	Explicit consent required before any biometric data is captured. Consent must specify modality, resolution, and purpose. Blanket consent is prohibited.
2	**STORAGE**	Individual designates storage jurisdiction, encryption standard, and retention duration. Default: encrypted at rest, individual's home GSSG tier jurisdiction, 10-year retention with mandatory re-consent.
3	**PROCESSING**	Each computational operation on biometric data requires purpose-specific authorization. Processing logs are immutable and auditable via CBGMODD provenance chains.
4	**MATCHING**	Biometric verification events (identity confirmation, access authentication) are governed by a separate consent layer from raw data processing. Matching produces a binary result; raw data is never transmitted to the verifying party.
5	**SHARING**	Cross-system or cross-jurisdictional transmission of biometric data requires explicit, granular, time-bounded consent. Each sharing event is logged on Gracechain as a trust transaction.
6	**DERIVATION**	Creation of derivative biometric products (templates, hashes, behavioral models) requires separate consent from raw data

		collection. Derivatives inherit the consent constraints of their source data.
7	**DELETION**	Individual retains irrevocable right to demand complete deletion of any or all biometric data, including derivatives. Deletion is verified through CBGMODD proof-of-erasure protocol and logged on Gracechain.

4.2 Anti-Surveillance Provision

FAVORS includes an explicit anti-surveillance provision: no entity operating within the NAC framework—governmental, corporate, or institutional—may conduct passive biometric surveillance (the capture of biometric data from individuals who have not provided explicit, informed consent for the specific surveillance context). This provision is absolute and non-waivable. Emergency security provisions under the Resolution Protocol (ERES-RESPROTOCOL-2026-001, Section 6) may compress procedural timelines but may not suspend the anti-surveillance provision.

5. CBGMODD: DATA-CHAIN INTEGRITY AND PROVENANCE

CBGMODD is the data integrity subsystem of NAC. It provides cryptographically verifiable provenance chains for every datum that enters, traverses, or exits any NAC-governed system. Where FAVORS governs consent over biometric data specifically, CBGMODD governs the integrity and traceability of all data universally.

5.1 Provenance Chain Architecture

Every data object within the NAC framework carries an immutable provenance chain—a cryptographically signed record of its origin, every transformation it has undergone, every system that has accessed it, and every consent authorization under which those accesses occurred. The provenance chain serves as the data object's "biography": a complete, tamper-evident account of its life within the system.

Provenance chains are structured as directed acyclic graphs (DAGs) in which each node represents a data state and each edge represents a transformation or access event. The chain is signed at each transition using the acting entity's IDIPITIS credentials, creating a multi-party attestation trail that would require collusion across multiple independent credential holders to falsify.

5.2 Proof-of-Erasure

A critical innovation within CBGMODD is the proof-of-erasure protocol, which addresses one of the most persistent challenges in data privacy: verifying that deleted data is actually gone. When an individual exercises their FAVORS deletion right (Layer 7),

CBGMODD generates a cryptographic proof that the specified data has been irreversibly removed from all storage locations, including backups, caches, and derivative products. The proof-of-erasure is logged on Gracechain and is independently auditable.

5.3 CBGMODD and the Public Reasoning Archive

CBGMODD provides the integrity backbone for the Public Reasoning Archive specified in the Resolution Protocol. Every entry in the Archive carries a full provenance chain, ensuring that the deliberative record cannot be retroactively altered, selectively redacted, or falsely attributed. This is the technical instantiation of the ERES epistemological principle that shared reasoning constitutes shared wealth—CBGMODD ensures that the wealth, once shared, cannot be stolen or corrupted.

6. GRACECHAIN: TRUST LEDGER AND MERITOCRATIC CREDENTIALING

Gracechain is the distributed trust ledger of the NAC framework. It is not a financial ledger (that function is served by Meritcoin); rather, Gracechain records, validates, and makes auditable the trust transactions that constitute the social fabric of NAC governance. Every credential earned, every consent given, every resolution honored, every commitment fulfilled—these are trust events, and Gracechain is their permanent, transparent record.

6.1 Trust Transaction Taxonomy

Gracechain recognizes four categories of trust transaction:

- **Credential Events.** Issuance, renewal, suspension, or revocation of SCALULAR certifications, IDIPITIS clearance levels, and EarnedPath milestones. Each event is signed by the credentialing authority and the credential holder.
- **Consent Events.** Granting, modification, or revocation of FAVORS consent at any of the seven layers. Consent events are the most privacy-sensitive trust transactions and are recorded using zero-knowledge proofs that confirm the consent action without revealing the underlying biometric data.
- **Adjudicative Events.** Resolution Protocol outcomes, appeals, remands, and compliance determinations. These events link to the Public Reasoning Archive via CBGMODD provenance chains.
- **Institutional Transparency Events.** Mandatory disclosures by NAC-governed institutions: algorithmic audit results, BERA index publications, resource allocation reports, and governance decisions. Institutional transparency is not voluntary; it is a structural requirement enforced by Gracechain's architecture.

6.2 Trust Score and the RCI

Gracechain maintains a rolling trust score for every entity (individual and institutional) within the NAC framework. The trust score is a composite of credential weight, consent compliance history, adjudicative outcome adherence, and transparency record. It feeds directly into the BERA Resonant Continuity Index (RCI = P_Ω_norm × ARI_sys × VibConst), where P_Ω_norm (normalized probability of systemic continuity) is substantially influenced by the aggregate Gracechain trust scores of the entities within the measured system.

7. SECUIR: INFRASTRUCTURE PROTECTION

SECUIR is the physical and digital infrastructure protection subsystem of NAC. While IDIPITIS governs identity, FAVORS governs biometric data, CBGMODD governs data integrity, and Gracechain governs trust, SECUIR governs the material substrate upon which all of these depend. A system's security is ultimately bounded by the integrity of its physical and network infrastructure; SECUIR ensures that this substrate remains resilient.

7.1 Defense-in-Depth Architecture

SECUIR implements a five-ring defense-in-depth architecture:

- **Ring 1 — Perimeter.** Physical facility security, network boundary protection, and electromagnetic shielding for sensitive installations. Governed by SSPS (Solid-State Smart-City Protection) certifications within SCALULAR.
- **Ring 2 — Transport.** Encrypted communication channels, secure routing protocols, and tamper-evident data transmission across all SOMT/GAIA coordination networks.
- **Ring 3 — Compute.** Hardened processing environments for BERA calculations, FAVORS biometric matching, IDIPITIS clearance evaluations, and Gracechain consensus operations. Compute environments are airgapped from public networks for Tier 5+ operations.
- **Ring 4 — Storage.** Encrypted-at-rest data repositories with geographically distributed redundancy. Storage integrity is continuously verified through CBGMODD provenance chain audits.
- **Ring 5 — Energy.** VERTECA-governed energy supply resilience. All critical NAC infrastructure operates on redundant energy sources with minimum 72-hour autonomous operation capacity. Energy grid attacks (Threat Class IV) are countered through VERTECA's distributed generation and storage architecture.

7.2 SECUIR and VERTECA Integration

The SECUIR-VERTECA integration point is among the most critical in the entire SPT architecture. A governance system that loses power loses everything—security boundaries collapse, biometric verification ceases, data integrity monitoring halts, and trust ledger consensus fails. VERTECA's Vertical Energy Resonance Temporal

Environmental Cybernetic Architecture provides the energy resilience that makes the rest of the SPT possible. SECUIR treats energy as a first-class security concern, not an operational convenience.

8. NBERS: NATIONAL BIOMETRIC ENROLLMENT AND RIGHTS SCAFFOLDING

NBERS is the enrollment and onboarding subsystem that creates the initial bridge between an individual and the NAC security-privacy-trust architecture. It is the “front door” through which every person enters the system, and its design carries extraordinary ethical weight: the enrollment experience establishes the individual’s first impression of whether the system respects their sovereignty or demands their submission.

8.1 Enrollment Principles

NBERS enrollment is governed by five non-negotiable principles:

4. **Informed Voluntarism.** Enrollment must be genuinely voluntary. No government benefit, employment opportunity, or social service may be conditioned on NAC enrollment. UBIMIA provisions are designed to function in parallel with legacy systems during transition periods to ensure no one is coerced by economic necessity.
5. **Comprehensible Consent.** The enrollment consent process must be conducted in the individual’s primary language, at a reading level appropriate to their educational background, with mandatory opportunity for questions and withdrawal. Consent obtained through incomprehension is void.
6. **Minimum Viable Collection.** NBERS collects only the biometric and identity data strictly necessary for Tier 1 SSSC credentialing. Additional data collection is deferred to later lifecycle stages and governed by progressive FAVORS consent layers.
7. **Immediate Rights Activation.** Upon completion of enrollment, the individual immediately receives their full Tier 1 credential set and access to all UBIMIA baseline provisions. There is no probationary period, no “pending” status, and no delayed activation.
8. **Reversibility.** An individual may withdraw from the NAC system at any time by exercising FAVORS Layer 7 (Deletion) rights across all their data. Withdrawal triggers CBGMODD proof-of-erasure and Gracechain archival logging. The system may not retain any data from a withdrawn individual except the anonymized fact that a withdrawal occurred (for demographic integrity of aggregate statistics).

9. DEGRADED STATE MANAGEMENT AND REMEDIATION

When any component of the SPT Triad Integrity Condition (Section 1.4) fails, the system enters a formally defined degraded state. Degraded states are not emergencies by default; they are signals—cybernetic feedback indicating that the security-privacy-trust equilibrium has been disturbed and requires corrective action. The SPT specification defines three levels of degradation:

9.1 Degradation Levels

Level	Designation	Trigger and Response
AMBER	**DRIFT**	One SPT index approaches its threshold boundary (within 10% of floor). Automated alerts to relevant GSSG tier governance body. Enhanced monitoring activated. No operational restrictions imposed. Remediation timeline: 30 days.
ORANGE	**BREACH**	One or more SPT indices have crossed their threshold. Active remediation mandated. Affected subsystems enter enhanced audit mode. Resolution Protocol emergency provisions may be invoked. Remediation timeline: 14 days.
RED	**COLLAPSE**	Multiple SPT indices have failed simultaneously or a single index has fallen below 50% of baseline. Systemic integrity is compromised. Immediate escalation to the next GSSG tier. All affected operations enter safe-mode. Planetary Reconciliation Council notified if Tier 4+ systems are affected. Remediation timeline: immediate.

9.2 Remediation Protocol

Remediation follows a four-step cycle derived from the cybernetic feedback principles embedded in the Triune Math:

9. **Diagnose.** Identify which SPT component(s) have degraded, which threat class(es) are implicated, and which subsystems (IDIPITIS, FAVORS, CBGMODD, Gracechain, SECUIR, NBERS) are affected. MIEVM validation is invoked for complex or ambiguous diagnoses.
10. **Contain.** Isolate the affected domain to prevent cascading degradation. Containment actions are evaluated against Triune Math constraints: $C = R \times P/M$ ensures that containment resources are proportionate to purpose and method.

11. **Restore.** Return affected indices to their baseline values. Restoration actions must satisfy the BERA threshold requirements specified in the Resolution Protocol. Partial restoration is acceptable as an interim measure provided a full restoration plan with timeline is filed with the applicable GSSG tier body.
12. **Harden.** Implement structural changes to prevent recurrence. Hardening actions are evaluated against REAL = (E·M·R)/(T·S) to ensure they are implementable within the spatiotemporal constraints of the affected domain. Hardening results are logged on Gracechain and incorporated into the Public Reasoning Archive.

10. CROSS-SYSTEM DEPENDENCY GRAPH

The six SPT subsystems do not operate in isolation; they form a tightly coupled dependency graph in which the failure of any one subsystem affects the operational capacity of all others. Understanding this dependency structure is essential for effective threat response and system design.

The critical dependency relationships are:

- **IDIPITIS depends on FAVORS** for biometric verification of identity claims. Without FAVORS, identity cannot be reliably authenticated.
- **FAVORS depends on CBGMODD** for provenance tracking of biometric data and proof-of-erasure capability. Without CBGMODD, consent enforcement is unverifiable.
- **CBGMODD depends on Gracechain** for trust anchoring of provenance chain signatures. Without Gracechain, provenance attestations lack a trust root.
- **Gracechain depends on SECUIR** for infrastructure integrity of the distributed ledger. Without SECUIR, the trust ledger is vulnerable to manipulation.
- **SECUIR depends on VERTECA** for energy resilience. Without VERTECA, infrastructure protection collapses during power disruptions.
- **NBERS depends on all five** preceding subsystems for enrollment integrity. NBERS is the integration point where all SPT subsystems converge to create the individual's entry into the system.

This circular dependency structure is deliberate. It means that no single subsystem can be compromised in isolation without producing detectable effects across the entire SPT architecture—a design property that converts the apparent vulnerability of tight coupling into a strength: comprehensive intrusion detection through systemic resonance.

11. MIEVM META-VALIDATION OF THE SPT ARCHITECTURE

Consistent with ERES research methodology, this specification has been and will continue to be subjected to Multi-Instrument Ensemble Validation (MIEVM). The SPT

architecture—as the protective core of the entire NAC framework—demands the highest epistemic rigor. MIEVM validation of the SPT employs multiple independent analytical instruments (including AI systems operating as parallel validation nodes) to stress-test the specification against adversarial scenarios, identify logical inconsistencies, and validate the cross-system dependency graph against failure-mode analysis.

MIEVM validation is not a one-time event. The SPT specification mandates ongoing ensemble validation at six-month intervals, with ad hoc validation triggered by any ORANGE or RED degradation event. Validation reports are appended to the Public Reasoning Archive and are subject to CBGMODD provenance tracking.

12. CLOSING DECLARATION

This specification recognizes a truth that most security architectures prefer to ignore: that security without privacy is oppression, that privacy without trust is isolation, and that trust without security is naïveté. The three must be held together, in dynamic tension, as a single system—a triad whose integrity is the precondition for every other governance function within the NAC framework.

The SPT architecture is offered not as a finished product but as a formal draft—an architectural declaration of intent that maps the full scope of the protective challenge and proposes a unified response. Implementation details, cryptographic specifications, performance benchmarks, and operational procedures remain to be developed in subsequent annexes. What this document establishes is the shape of the solution: the subsystems, their relationships, their shared threat model, and the moral vector that governs their design.

That moral vector bears repeating: the purpose of security, privacy, and trust within the NAC framework is not to control people but to liberate them. To build a world in which individuals possess genuine sovereignty over their identities, their data, and their participation in governance. To ensure that institutions earn trust rather than demand it. And to do all of this in service of the enduring imperative: ***Don’t hurt yourself. Don’t hurt others. Build for generations to come.***

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